



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Theory of Computation / Theory of Computing
2	Course Code	CSE 2233 / CSI 233
3	Trimester and Year	Fall 2025
4	Pre-requisites	Algorithms, CSI-227, Algorithms Laboratory, CSI-228
5	Credit Hours	3.0
6	Section	
7	Class Hours	
8	Class Room	
9	Instructor's Name	Asif Ahmed Utsa
10	Contact Info	Email: asif@cse.uiu.ac.bd
11	Office	Room: 419-A
12	Counseling Hours	Please take a look at the list of CNH hour on the door of 419-A
13	Text Book	<i>Introduction to Automata Theory, Languages, and Computation</i> , by John Hopcroft, Rajeev Motwani, and Jeffrey Ullman
14	Reference	1. http://infolab.stanford.edu/~ullman/ialc.html 2. Introduction to the Theory of Computation (2nd ed. 2006). by Michael Sipser 3. Languages and Machines: An Introduction to the Theory of Computer Science.(3rd ed), by Thomas Sudkamp
15	Course Contents (approved by UGC)	Basic concepts of Finite Automata and Languages: Alphabet, String, Language; Finite Automata: Deterministic Finite Automata, Nondeterministic Finite Automata, Equivalence of Deterministic and Nondeterministic Finite Automata, Finite Automata With Epsilon Transitions, Minimization of Automata; Regular Expressions: Regular Expression & it's Application, Equivalence of Finite Automata and Regular Expressions; Context-Free Grammars: Application of CFG, Parse Tree, Ambiguity in Grammars and Languages, Normal Forms for Context-Free Grammars; Pushdown Automaton: Languages of a PDA,

		Equivalence of PDA's and CFG's, Deterministic Pushdown Automata; Introduction to Turing Machine; Intractable Problems: NP completeness.																																																																													
16	Course Outcomes (COs)	<table><tr><th>COs</th><th>Description</th></tr><tr><td>CO1</td><td>Acquire a fundamental understanding of the core concepts in automata theory and formal languages.</td></tr><tr><td>CO2</td><td>An ability to design grammars, regular expression and automata for different language classes.</td></tr><tr><td>CO3</td><td>An ability to identify formal language classes and its properties.</td></tr><tr><td>CO4</td><td>Acquire a fundamental understanding of core concepts relating to the theory of computation and computational models including decidability and intractability.</td></tr></table>	COs	Description	CO1	Acquire a fundamental understanding of the core concepts in automata theory and formal languages.	CO2	An ability to design grammars, regular expression and automata for different language classes.	CO3	An ability to identify formal language classes and its properties.	CO4	Acquire a fundamental understanding of core concepts relating to the theory of computation and computational models including decidability and intractability.																																																																			
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18	CO with Assessment Methods	<table><tr><th>CO</th><th>Assessment Method</th><th>(%)</th></tr><tr><td>-</td><td>Attendance</td><td>5</td></tr><tr><td>-</td><td>Assignments</td><td>5</td></tr><tr><td>-</td><td>Class Tests</td><td>20</td></tr><tr><td>1, 3</td><td>Midterm exam</td><td>30</td></tr><tr><td>2, 3, 4</td><td>Final exam</td><td>40</td></tr></table>	CO	Assessment Method	(%)	-	Attendance	5	-	Assignments	5	-	Class Tests	20	1, 3	Midterm exam	30	2, 3, 4	Final exam	40																																																											
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Class	Topics/Assignments	COs	Reading Reference	Lecture Outcomes/Activities
1	Basic concepts of finite automata, Languages, Alphabet, String	1	Ch1	Lecture, Q/A
2	Deterministic Finite Automata	1	Ch1	Lecture, Q/A
3, 4	Deterministic Finite Automata	1, 2	Ch2	Lecture
5	Deterministic Finite Automata	1, 2	Ch2	Lecture
6	Nondeterministic Finite Automata	3	Ch2	Lecture, Class Test
7	Nondeterministic Finite Automata	1, 2	Ch2	Lecture, Q/A, HW
8, 9	NFA to DFA Conversion	2	Ch3	Lecture,
10, 11	e-NFA to DFA Conversion	2	Ch3	Lecture, Class Test
12	Regular Expression	3	Ch4	Lecture
	MIDTERM EXAM			
13	Context Free Grammars	2	Ch5	Lecture
14	Context Free Grammars	2	Ch5	Lecture
15	Context Free Grammars	2	Ch5	Lecture
16	Parse Tree and Derivation	2	Ch5	Lecture
17	Ambiguity in Grammars and Languages	2, 3	Ch5	Lecture, Quiz
18	Definition of the Pushdown Automaton	4	Ch6	Lecture, Class Test
19	Design of Pushdown Automaton	3, 4	Ch6	Lecture, Q/A
20	Design of Pushdown Automaton	4	Ch6	Lecture, HW, Q/A
21	Introduction to Turing Machine	4	Ch8	Lecture, HW, Q/A
22	Design of Turing Machine	3	Ch8	Lecture, Class Test
23	Design of Turing Machine	4	Ch8	Lecture,
24	Overview	4		Lecture, Q/A

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	5%
Assignments	5%
Class Tests	20%
Mid Term	30%
Final Exam	40%

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks %	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
P01	An ability to apply knowledge of mathematics, science, and engineering
P02	An ability to identify, formulate, and solve complex engineering problems
P03	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
P04	An ability to investigate complex problems using research-based knowledge and research methods design and conduct experiments, as well as to analyze and interpret data
P05	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
P06	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
P07	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts
P08	An understanding of professional and ethical responsibility
P09	An ability function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
P010	An ability to communicate effectively
P011	Project management and finance
P012	A recognition of the need for, and an ability to engage in life-long learning